Exercise 1

Describe the outcome, including the qualitative behaviour of the left and right masses after collision.

The particles approach each-other head on (but off-center), and the leftmost particle is travelling twice as fast. When the particles hit each other, the rightmost one begins to travel toward the bottom right, and the left one travels to the upper left. The particle that was initially travelling more slowly becomes faster, which is a manifestation of conservation of linear momentum, which is directed rightward.

Compute the initial and final momentum, energy and angular momentum before and after this collision.

```
Conserved quantities [before collision] (about LVector3f(0, 0, 0)):
  P = (1.0, 0.0, 0.0)
  L = (0.0, 1.2000000476837158, -25.0)
  K = 2.5

Conserved quantities [after collision] (about LVector3f(0, 0, 0)):
  P = (1.0, 0.0, 0.0)
  L = (0.0, 1.218420147895813, -25.0)
  K = 2.500000238418579
```

Exercise 2

What is the inverse mass of the 3rd object, and why does this make sense here?

The inverse mass is zero. It makes sense because when the acceleration is computed via Newton's Third Law (a = F * inverse_mass), if inverse_mass = 0, then the acceleration is zero, no matter how hard a force is being imparted.

Describe what happens. What are the final velocities of all three masses?

The basketball hits the wall, and at nearly the same instant, the basketball hits the ping-pong ball. The basketball's momentum is flipped and remains almost the same in magnitude, but loses some momentum. When the ping-pong ball hits the basketball, it goes flying back in the opposite direction a lot faster than it approached. The final velocities are listed below:

Velocities [after 2nd collision]:
p0: v = (-29.603954315185547, 0.0, 0.0)
p1: v = (-9.60395622253418, 0.0, 0.0)
p2: v = (0.0, 0.0, 0.0)

Describe the sequence of events that is occurring to achieve this result.

1. Basketball hits the wall; velocity/momentum flips
2. Basketball hits the ping-pong ball; the basketball slows down a little bit
3. The ping-pong ball, because it's so small in mass, accelerates greatly in the opposite direction to conserve momentum in the second ball-ball collision

Modify the set-up in minor ways (e.g. initial x positions, initial vx for the two masses). Always keep the 3 masses in row in the same order with no initial overlap and always with the left two masses moving at the same speed. How does this affect the outcome?

Starting positions don't change the final velocities; the same two collisions happen regardless (ball-wall, then ball-ball).

Starting velocity of two balls	*Final velocity (ping pong, basketball)*
10	-29.603954315185547, -9.60395622253418
20	-59.207908630371094, -19.20791244506836
40	-118.41584777832031, -38.41584014892578
80	-236.83169555664062, -76.83168029785156

From the above table, it appears that the final velocity of the balls scales proportionately with the starting velocity. When the start velocity is doubled, the final velocity is approximately doubled.

What would be the limiting result if the first mass was made very light (without changing the radius)? You can experiment with your code to explore this. Express your answers for the three final velocities in terms of multiples of the initial vx . That is – give an answer that would apply for any choice of vx > 0.

We will use a starting velocity of 10

<u>Inverse mass of a ping-pong ball</u>	*<u>Final velocity Relative to Initial (Ping Pong Ball)</u>*	*<u>Final velocity Relative to Initial (Basketball)</u>*
1	-29.603954315185547 / 10 = -2.96	-9.60395622253418 / 10 = -0.9603
10	-29.96002960205078 / 10 = -2.996	-9.960036277770996 / 10 = -0.9600
1000	-29.999591827392578 / 10 = -2.9999	-9.99959659576416 / 10 = -0.9996
1e9	-29.99999237060547 / 10 = -2.9999	-9.999996185302734 / 10 = -0.999996

The limiting result for the ping-pong ball is that final_v → -3 * initial_v;
For the basketball, it's final_v → -1 * initial_v

Exercise 3

Print out the conserved quantities at the start and after 30 seconds and report those in your hand-in pdf.

Conserved quantities [start] (about LVector3f(0, 0, 0)):
P = (30.071762084960938, 0.0, -5.5964202880859375)
L = (-559.6420288085938, -123.84687042236328, -3007.17626953125)
K = 439.94279766082764

Conserved quantities [after 30s] (about LVector3f(0, 0, 0)):
P = (-3.6845626831054688, 0.0, 29.88220977783203)
L = (2988.22119140625, 273.23541259765625, 368.456298828125)
K = 439.9428367614746

What is conserved now? If something is not conserved, explain why not.

Energy is conserved because we are forcing only elastic collisions to occur. Momentum is not concerned, however, due to wall collisions. When a particular ball collides with a wall, the component of velocity perpendicular to the wall gets flipped, which also flips that component of momentum. In this case, the energy is conserved, but momentum is certainly not. This is why linear and angular momentum are not conserved.